Appendix K - Block Diagrams

Schematic-level layout of the three internal buses that connect the chips inside the Intuition Engine: the video compositor, the audio mixer, and the main system bus. Each diagram is drawn at the granularity a programmer needs to reason about cross-chip interactions, not at the granularity an engineer would use to fabricate the silicon.

K.1 The video compositor

The compositor has six layers. Each layer comes from one of the six video engines. Layers are listed in back-to-front order; a higher layer covers the lower ones where it draws non-transparent pixels.

```
                  +------------------------------+
                  |  HDMI / display output       |
                  +--------------^---------------+
                                 |
                  +--------------+---------------+
                  |   Video compositor           |
                  |   (per-pixel layer mux,      |
                  |    palette, blend modes)     |
                  +-+----+----+----+----+----+--+
                    |    |    |    |    |    |
         layer 20   |    |    |    |    |    |   (front)
                    |    |    |    |    |    +-- Voodoo 3D
                    |    |    |    |    +------- ULA
                    |    |    |    +------------ ANTIC + GTIA
                    |    |    +----------------- TED video
                    |    +---------------------- VGA
                    +--------------------------- VideoChip (layer 0, back)
```

Each engine writes into its own framebuffer (in main VRAM or in its private aperture); the compositor reads all six on every output line and resolves them into a single stream of RGBA pixels.

All six sources are collected on the same `60` Hz frame cadence. A chip that has scanline work still completes that work in its own frame before the compositor resolves the stack.

K.2 The audio mixer

The audio mixer has separate engine inputs, plus the SFX channels, and one stereo output stream.

```
SoundChip --+
PSG / AY --+
SN76489 ---+
SID  ------+
SID2 ------+
SID3 ------+
POKEY -----+--->  Mixer  --->  Overdrive ---> Filter ---> Reverb ---> Output
TED audio -+      (sum,         (global)      (global)    (global)
AHX -------+       per-chip      voice)
MIDI/MUS --+       gain,
Live MIDI -+       per-chip
MOD -------+       mute)
WAV -------+
Paula DMA -+
           |
SFX ch 0-31 +
```

The SoundChip's own filter, the SID family's resonant filter, and the engine-internal effects all feed the mix before the global overdrive / filter / reverb stage; the global effects apply once per output sample to the summed signal.

MIDI/MUS and Live MIDI share the RawlandMini synth and voice pool. The diagram shows both control surfaces because one starts stored song data and the other streams immediate MIDI bytes.

K.3 The system bus

Every CPU and every device hangs off one shared `64`-bit physical bus. The bus carries `64`-bit physical addresses and supports `8`, `16`, `32`, and `64`-bit transfers through CPU and device adapters. It arbitrates simultaneous accesses on a round-robin basis; there is no priority encoding above the CPU / DMA distinction.

```
                  +--------------------+
                  |    Main RAM        |
                  +---------^----------+
                            |
                            |
    +------+   +------+   +--+--+   +------+   +------+   +------+
    | IE64 |   | IE32 |   | M68K|   | 6502 |   | Z80  |   | x86  |
    +--+---+   +--+---+   +--+--+   +--+---+   +--+---+   +--+---+
       |          |          |         |          |          |
       +----------+----------+--+------+----------+----------+
                              |
                  +-----------+----------+
                  |   System bus         |
                  |   (64-bit address,   |
                  |    8/16/32/64-bit    |
                  |    transfers,        |
                  |    arbitrated)       |
                  +-+---+---+---+---+---+--+
                    |   |   |   |   |   |
              MMIO  |   |   |   |   |   |  DMA engines:
              regs  |   |   |   |   |   |   - blitter
                    |   |   |   |   |   |   - Amiga Paula DMA
              VGA --+   |   |   |   |   +-- - SFX sample DMA
              Video chip|   |   |   |       - copper
              ULA ------+   |   |   |
              Audio --------+   |   |
              File I/O ---------+   |
              Coprocessor ---------+
```

IE32, M68K, and x86 are 32-bit bus clients. The 8-bit CPUs (6502, Z80) reach the bus through an address translator that turns their 16-bit address space into selected low-window bus addresses, with the bank registers described in Chapters 27 and 28 selecting which translation applies.

K.4 Coprocessor channels

The coprocessor block (Chapter 32) is a many-to-many channel between the main CPU and a pool of worker instances. Each running instance listens on its own ring buffer; the main CPU posts work items into that ring and reads the result back.

```
    Main CPU
    |
    |  COSTART / COCALL  -->  +-------------------+
    |                         |  Assigned ring    |
    |  COSTATUS / COWAIT  <-- |  (per instance)   |
    |                         +---------+---------+
    |                                   |
    |                         +---------+---------+
    |                         |  Worker instance  |
    |                         |  type plus number |
    |                         +-------------------+
```

Six worker types share the protocol: an IE64 worker, an IE32 worker, a 6502 worker, a Z80 worker, an M68K worker, and an x86 worker. The main CPU is whichever one boots the machine; all the others are available as workers when not booted. M68K, x86, and IE64 provide instances `0` and `1`; IE32, 6502, and Z80 provide instance `0` only.

The mailbox reserves twelve ring slots from `$790000` to `$792FFF`. Each slot has a uniform `$400` stride, and its index is `cpuTypeIndex * 2 + instance`. The unused second slots for IE32, 6502, and Z80 remain reserved so every supported worker follows the same address rule.

K.5 Bus translation for the small CPUs

The 6502 and Z80 see the same overall bus through a 16-bit-to- 64-bit translator. In practice their documented apertures target the low memory and MMIO window. The translator has two independent functions:

```
6502 / Z80 address (16-bit)
     |
     v
+----+-----------------------+
| Decode page                |
|  - bank registers $F7xx? --> intercepted, never reach bus
|  - MMIO mirror $F000-$FFF9? -> +$F0000 -> bus
|  - $E000-$EFFF banked window-> + bank * 4 KB -> bus
|  - everything else --------- > main RAM page
+----+-----------------------+
     |
     v
bus address (64-bit, low-window value here)
```

The bank registers at `$F700-$F705` (low/hi pairs for apertures 1, 2, 3) and `$F7F0` (aperture 0) are captured before the translator runs. A write to `$F700` does not reach `$F0700`; it loads the low byte of bank register 1.